# Trapped by Cash

## A Theory of Voluntary Informality Among the Economically Non-Viable

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# Part I

## The Puzzle

$\sim$10 minutes

# The Taquero's Paradox

Consider a **taquero**—food stall owner—working the same corner for 10 years:

- Neighbors are customers (loyalty)
- Supplier extends 30-day credit
- Landlord hasn't raised rent in 3 years

**Survey asks:** "How much did you earn?"

He reports: **Cash in − Cash out = Positive**

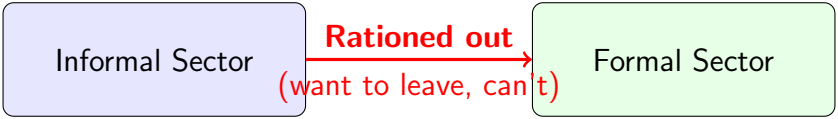
**But price his relationships at market...**

- His time at shadow wage
- Family labor at shadow wage
- Equipment depreciation

$\Rightarrow$ **Economic profit** $< 0$

**Yet he won't leave.** Why?

# The Standard View: Exclusion




**Policy implication:** Raise formal offer rate $\rho$

- Job matching services
- Hiring subsidies
- Training programs

**Evidence:** Mixed at best (Card et al. 2018)

# This Paper's View: The Cash Flow Veil

**The mechanism:**

- Owners observe **cash flow** ($CF$), not economic profit ($\pi_{adj}$)
- $CF > 0$ even when $\pi_{adj} < 0$
- If $CF \geq w^*$ (formal wage), they **decline** formal offers

**Policy implication:** Raising $\rho$ is **irrelevant** for these workers

# The Central Finding

**43%**

of informal firms are **loss-making**
(after imputing owner labor + depreciation)

**Yet 97% have positive cash flow**

# The Four-Type Partition

$\pi_{adj} > 0$ (Viable)

| | $CF < w^*$ | $CF \geq w^*$ |
|---|---|---|
| $\pi_{adj} > 0$ (Viable) | **Type B**<br>Viable + Rationed<br>n=2 (0.4%) | **Type A**<br>Viable + Trapped<br>n=284 (56.8%) |
| $\pi_{adj} \leq 0$ (Loss-making) | **Type D**<br>Loss + Rationed<br>n=64 (12.8%) | **Type C**<br>Loss + Trapped<br>n=150 (30.0%) |

This is the population the paper is about

$\pi_{adj} \leq 0$ (Loss-making)

# The Policy Mistake

**Standard ALMP assumes:**

- All loss-makers are Type D
- They want formal jobs
- They're rationed out
- $\Rightarrow$ Raise $\rho$

**The reality:**

- 70% of loss-makers are Type C
- They would **decline** formal offers
- Raising $\rho$ does nothing

**Mistargeting Ratio**

$$\frac{\text{Type C}}{\text{Type D}} = \frac{160}{54} = \mathbf{2.96}$$

For every worker correctly targeted, nearly 3 receive resources that cannot affect their outcome.

# Part II

## Literature & Contribution

$\sim$8 minutes

# Five Literature Streams

| Literature | Key Papers | This Paper |
|---|---|---|
| Dual view debate | La Porta & Shleifer (2014), Maloney (2004), Ulyssea (2020) | Micro-foundation for why "survivalists" don't exit |
| Structural models | Ulyssea (2018), Meghir et al. (2015) | First model with imperfect profit observation |
| Returns to capital | de Mel et al. (2008, 2012) | Reconcile high MPK with 43% loss-making |
| Financial diaries | de Mel et al. (2009), Collins et al. (2009) | Connect measurement to persistence |
| ALMPs | Card et al. (2018), Benhassine et al. (2018) | Explain modest effects via mistargeting |

# The Key Gap: No Model Has Imperfect Profit Observation

**Every structural model assumes firms observe $\pi$ exactly:**

- Ulyssea (2018, *AER*): firms draw productivity, observe profit, exit if $\pi < 0$
- Meghir, Narita, Robin (2015, *AER*): wage posting with known surplus
- Bobba, Flabbi, Levy (2022, *IER*): Nash bargaining over known profit

**But de Mel et al. (2009) show:**

- Correlation between self-reported and calculated profit: **0.2–0.3**
- 30% of firms report positive when calculated is negative

**This paper:** Replace $\pi$ with $CF$ in the exit condition

$$\text{Exit if } CF < 0 \quad (\text{not } \pi_{adj} < 0)$$

# Five Gaps This Paper Fills

1. **No structural model** incorporates imperfect profit observation
2. **No theory** connects profit mismeasurement to firm persistence
3. **Marginal $\neq$ average reconciliation** is unstated
   - High experimental returns (de Mel: 55%/year)
   - + 43% loss-making
   - = Not contradictory once you separate margin from average
4. **No micro-behavioral theory** of ALMP mistargeting
5. **No transaction-level decomposition** of the accounting gap

# Part III

## The Model

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# The Accounting Identity

**Two income concepts:**

**Conventional profit** (what owners observe):

$$\pi_{conv} = \text{Revenue} - \text{Explicit cash costs}$$

**Adjusted profit** (true economic profit):

$$\pi_{adj} = \pi_{conv} - \underbrace{w \cdot h^{own}}_{\text{imputed owner labor}} - \underbrace{w \cdot h^{family}}_{\text{imputed family labor}} - \underbrace{\delta \cdot K}_{\text{depreciation}}$$

**The accounting gap:**

$$\boxed{\pi_{conv} - \pi_{adj} = w(h^{own} + h^{family}) + \delta K}$$

$\Rightarrow$ **No cash changes hands** to produce the gap

# The Cash Flow Veil: Three Channels

### 1. Owner Labor

Owner works 50 hrs/week
At shadow wage: Bs 682/week
**No cash outflow**
Owner keeps residual

### 2. Family Labor

Spouse/children help
At shadow wage: Bs 340/week
**No cash outflow**
No wage paid

### 3. Depreciation

Equipment wears out
At 12%/year: Bs 92/week
**No cash outflow**
Until replacement

**Total invisible cost:** Bs 1,114/week (median)

$\Rightarrow$ Can have $\pi_{conv} > 0$ and $\pi_{adj} < 0$ simultaneously

# Dual Embedded Capital

Informal operation accumulates two forms of capital:

## Productive Capital ($\varepsilon_P$)

- Labor-augmenting knowledge
- Raises economic output
- Determines viability: $\pi_{adj} \gtrless 0$

## Relational Capital ($\varepsilon_X$)

- Loyal customers
- Supplier credit
- Below-market rent

Raises **cash flow** without raising economic profit

Determines trap: $CF \gtrless w^*$

**Key insight:** Both accumulate with tenure $\Rightarrow$ observationally indistinguishable from surveys

But economically distinct: $\varepsilon_P$ determines *if* viable; $\varepsilon_X$ determines *if* trapped

# The Trap Condition

[Trap Condition] An agent in informal self-employment declines a formal offer if and only if:

$$\boxed{CE(CF) \geq w^*}$$

where $CE(\cdot)$ is the certainty equivalent under CRRA utility.

This condition is **independent of**:

- Formal offer arrival rate $\rho$
- Job separation rate $\lambda$

**Policy implication:**

- $\rho$ is what ALMP raises $\Rightarrow$ irrelevant for trapped workers
- $\lambda$ is what employment protection targets $\Rightarrow$ also irrelevant

# Part IV

## Data

$\sim$8 minutes

# The Financial Diary

**Instrument:**

- 7-day double-entry diary
- Every transaction recorded
- Cash reconciliation daily
- September–October 2024

**Sample:**

- 500 informal firms
- 6 sectors
- 4 cities (La Paz/El Alto, Cochabamba, Santa Cruz, Tarija)
- 25,338 transactions

**Why diary, not survey?**

- Surveys: single question "monthly profit"
- Cannot separate $CF$ from $\pi_{adj}$
- Diary: construct both from transactions

**Validation:**

- Accounting identity closes to Bs 0.01
- 85% completion rate
- Cash balances reconcile daily

Full protocol: Hernani-Limarino (2026)
Data: `doi.org/10.7910/DVN/KZMHGU`

# Sample Characteristics

| | **Full Sample** (N=500) | **Viable** (N=286) | **Loss-making** (N=214) |
|---|---|---|---|
| Weekly revenue (Bs) | 2,074 | 2,632 | 1,327 |
| Conventional profit (Bs/wk) | 1,394 | 1,861 | 771 |
| Adjusted profit (Bs/wk) | 296 | 946 | −572 |
| Owner hours/week | 49.3 | 42.3 | **58.5** |
| Family hours/week | 24.9 | 15.3 | **37.7** |
| Tool value (Bs) | 39,806 | 44,642 | 33,342 |
| Years operating | 10.3 | 11.0 | 9.4 |
| Female owner (%) | 44.0 | 45.5 | 42.1 |
| Has debt (%) | 49.2 | 45.5 | 54.2 |

**Key pattern:** Loss-makers work **more hours** and use **more family labor**

$\Rightarrow$ Labor intensity drives the gap, not capital intensity

# Sector Composition

| Sector | N | Loss % | Type C % | Female % |
|---|---|---|---|---|
| Transport | 114 | 41.2 | **93.6** | 0.9 |
| Food Service | 54 | 59.3 | 68.8 | 83.3 |
| Commerce | 182 | 44.5 | 65.4 | 71.4 |
| Construction | 50 | 44.0 | 59.1 | 0.0 |
| Manufacturing | 67 | 28.4 | 57.9 | 43.3 |
| Prof. Services | 33 | 39.4 | **53.8** | 45.5 |

**Gradient matches theory:**

- Transport: high relational capital (routes, customers) $\Rightarrow$ highest Type C
- Prof. Services: portable human capital $\Rightarrow$ lowest Type C

# Part V

## Results

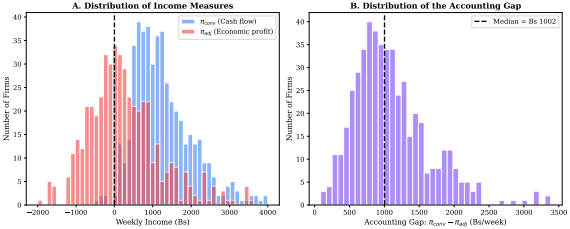
$\sim$15 minutes

# Result 1: The Accounting Gap

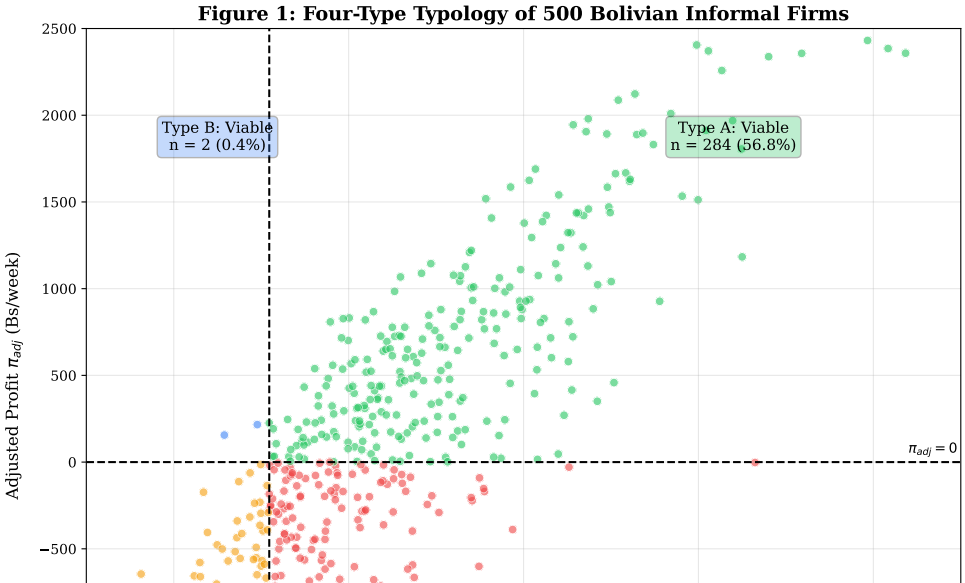
## Gap Decomposition

| Component | Median | Share |
|---|---|---|
| $\pi_{conv}$ | Bs 1,149 | — |
| Owner labor | Bs 655 | 61% |
| Family labor | Bs 257 | 31% |
| Depreciation | Bs 120 | 12% |
| **Gap** | **Bs 1,002** | 100% |
| $\pi_{adj}$ | Bs 140 | — |

**92% of gap is imputed labor**
(Owner 61% + Family 31%)



# Result 2: The Four-Type Distribution

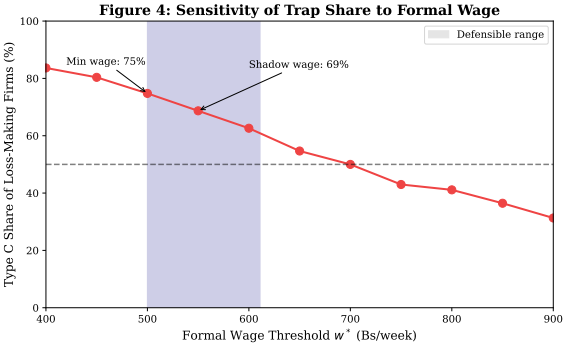


# Result 3: Robustness to $w^*$

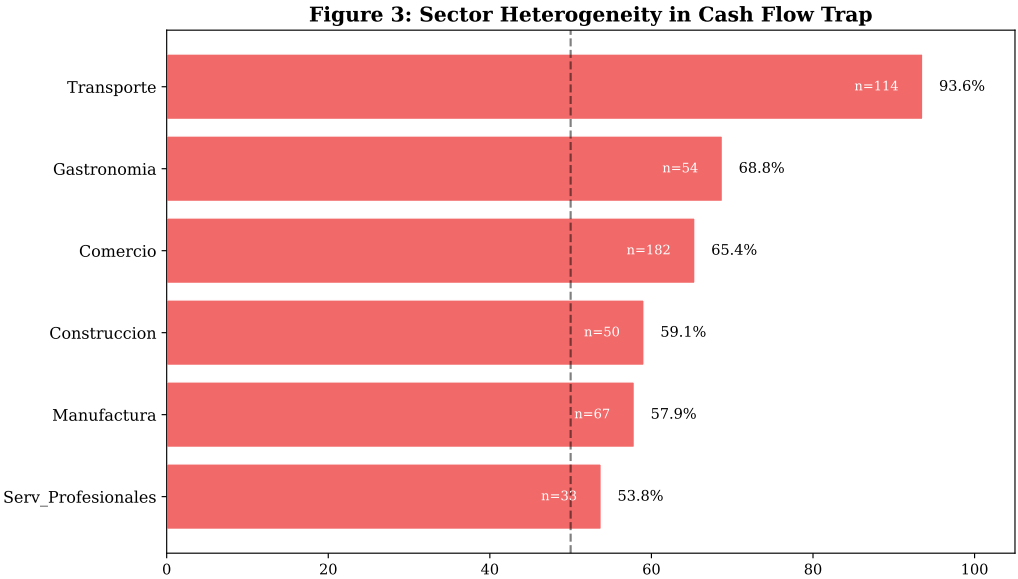
| $w^*$ | Type C | Type D | C/(C+D) |
|---|---|---|---|
| 500 (min wage) | 160 | 54 | **74.8%** |
| 546 (shadow) | 150 | 64 | **70.1%** |
| 611 (upper) | 132 | 82 | **61.7%** |
| 700 | 107 | 107 | 50.0% |
| 800 | 88 | 126 | 41.1% |

**Key:** Within defensible range (500–611), Type C share is always **62–75%**

Crosses 50% only at $w^* = 700$ (above any labor market benchmark)



# Result 4: Sector Heterogeneity

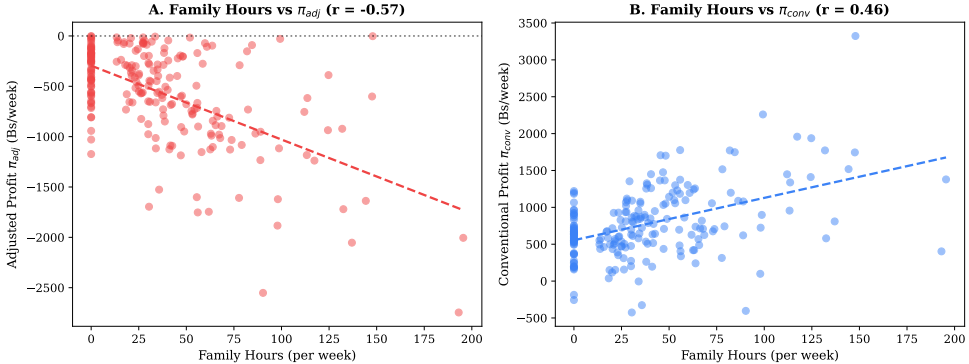


# Result 5: The Sign Reversal Test

**Prediction:** Family labor should...

- **Raise** cash flow (substitutes for hired labor)
- **Reduce** adjusted profit (adds imputed cost)

**Figure 5: Sign Reversal Test (Loss-Making Firms, n=214)**



# Result 6: Misclassification

**What would a standard survey conclude?**

| | Diary: Loss | Diary: Viable |
|---|---|---|
| **Survey: Loss** | 6 | 0 |
| **Survey: Viable** | 208 | 286 |

| Misclassification Rate | False Positive Rate |
|---|---|
| 41.6% | 42.1% |

$\Rightarrow$ Survey-based poverty targeting will **exclude** loss-making informal workers

# Part VI

## Policy Implications

$\sim$7 minutes

# Why ALMPs Show Modest Effects

**Card et al. (2018):**

- 200+ program evaluations
- Average short-run impact $\approx 0$
- Modest medium-run gains

**Benhassine et al. (2018):**

- Benin formalization experiment
- Zero profit gains
- Cost exceeded tax revenue

**This paper's explanation:**

Treatment effect is **diluted** across:

- Type D (12.8%): treatment works
- Type C (30.0%): treatment irrelevant

$$\text{Measured ATE} = \frac{D \cdot \tau_D + C \cdot 0}{D + C}$$

With $C = 2.96 \times D$, measured ATE is $\approx 25\%$ of true effect on Type D

# What Would Work for Type C?

1. **Financial literacy targeting imputed costs**
   - Drexler et al. (2014): rule-of-thumb training works
   - Rule: "Pay yourself a wage. Still have profit?"
2. **Severance packages for exit**
   - Lump sum to compensate for lost relational capital
   - Enable transition without cash flow shock
3. **Social insurance reform**
   - Contributory systems tax formality
   - Non-contributory alternatives reduce Type C mechanically
   - Levy (2008, 2018): Mexico's dual system subsidizes informality

# Implications for Poverty Measurement

**Problem:**

- Poverty targeting uses self-reported income
- Self-reported $= \pi_{conv}$ (cash flow)
- 43% are loss-making but report positive income
- $\Rightarrow$ Excluded from social protection

**Scale:**

- Bolivia: 85% informal employment (ILO 2024)
- If 43% are loss-making...
- $\approx$ 910,000 workers misclassified nationally
- Poverty rate understated by $\sim$20 pp in this population

# Part VII

## Conclusion

$\sim$2 minutes

# Summary

**The mechanism:**

- Owners observe cash flow, not economic profit
- Imputed labor + depreciation create gap (median Bs 1,002/week)
- 43% are loss-making; 97% have positive cash flow

**The result:**

- 62–75% of loss-makers satisfy trap condition ($CF \geq w^*$)
- They would decline formal offers despite negative profit
- ALMP mistargeting ratio: 2.96:1

**The implication:**

- Standard ALMP is largely irrelevant for loss-making informal workers
- Need interventions that reduce the veil or compensate for exit

# Replication

## Data & Code

**Harvard Dataverse**

https://doi.org/10.7910/DVN/KZMHGU

**GitHub**

https://github.com/wernerhl/ledgers-bolivia

**Companion Paper**

"Ledgers of the Self-Employed: Accounting for the Invisible Firm"

https://wernerhl.github.io/ledgers-bolivia

**Contact**

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# Thank You

Questions?

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# Appendix

# Shadow Wage Calculation

**Method:** Mincer regression on formal sector

$$\ln(w_i) = \alpha + \beta_1 \cdot \text{Educ}_i + \beta_2 \cdot \text{Exp}_i + \beta_3 \cdot \text{Exp}_i^2 + \varepsilon_i$$

**Result:** Shadow wage = Bs 13.64/hour

**Sensitivity:**

| Shadow wage | Loss-making % | Type C % |
|---|---|---|
| Bs 10.00 (40% of baseline) | 12.4% | 82.3% |
| Bs 13.64 (baseline) | 42.8% | 70.1% |
| Bs 16.00 (120% of baseline) | 52.6% | 63.4% |

**Key:** Type C share robust across shadow wage range

# Depreciation Rates

| Asset Type | Annual Rate | Source |
|---|---|---|
| Vehicles | 15% | Tax authority schedule |
| Machinery | 12% | Tax authority schedule |
| Equipment | 10% | Tax authority schedule |
| Furniture | 10% | Tax authority schedule |
| Structures | 5% | Engineering estimates |

**Method:** Straight-line depreciation on replacement value

**Depreciation share of gap:** 12% (median)
$\Rightarrow$ Labor dominates; depreciation is secondary

# Why Type B is Nearly Empty

**Type B:** Viable ($\pi_{adj} > 0$) but rationed ($CF < w^*$)

**Theoretical prediction:** Should be rare

**Why?**

- Requires $\pi_{adj} > 0$ but $CF < w^*$
- Since $CF = \pi_{adj} + wh + \delta K$ and all terms positive...
- Need $\pi_{adj}$ small AND $wh + \delta K$ small
- $\Rightarrow$ Low labor intensity AND low capital
- But low labor intensity $\Rightarrow$ low $\pi_{adj}$ (labor is the input)

**Empirical:** N = 2 (0.4%)
$\Rightarrow$ Model prediction confirmed